

2.3 Conditional Probability

Probabilities are calculated for a given sample space.

If we **restrict the sample space based on a given condition** and calculated the probability of an event for that restricted sample space, we get **conditional probability**.

	City			
	A	B	C	Total
Male	50	100	100	250
Female	50	25	75	150
Total	100	125	175	400

If a student is randomly selected, what is the probability that he/she is from **city A**?

Also, if a student is randomly selected, what is the probability that the student is **male**?

Now we will **restrict our sample space**.

If you select **male students only**, what is the probability that he is from city A?

If you select students **from A only**, what will be the probability that student is a male?

Definition 2.1: If A and B are any two events in a sample space S and $P(A) \neq 0$, the conditional probability of B given A is,

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

With reference to the previous example what is the probability that student is from city B given that the student is a female?

	City			
	A	B	C	Total
Male	50	100	100	250
Female	50	25	75	150
Total	100	125	175	400

	City			
	A	B	C	Total
Male	50	100	100	250
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Example: A local airline knows from past experience that the probability is 0.80 that a takeoff will be on time and it is 0.72 that takeoff will be on time and will also be landed on time. What is the probability that such a trip will be landed on time given that the takeoff happened on time?

Let T is the event that takeoff happened on time and L be the event that it landed on time.

Theorem 2.9:

If A and B are any two events in sample space S and $P(A) \neq 0$

$$P(A \cap B) = P(A) \cdot P(B|A)$$

This theorem is known as the multiplication rule.

Also by considering the fact that $A \cap B = B \cap A$

an alternative multiplication rule can be derived as,

$$P(A \cap B) = P(B) \cdot P(A|B)$$

Example: If we randomly select two television tubes in succession from a shipment of 240 television tubes of which 15 are defective. What is the probability that they will both be defective?

If we assume equal probabilities for each selection, the probability that the first tube will be defective = $\frac{15}{240}$

and the probability that the second tube will be defective given that the first tube is defective = $\frac{14}{239}$

(Assumed that we are sampling without replacement).

Thus the probability both tubes will be defective is

$$\frac{15}{240} \cdot \frac{14}{239} = \frac{7}{1912}$$

Example:

If $P(A)=0.5$, $P(B)=0.7$ and $P(A \cup B)=0.9$, find

$P(A|B)$

$P(B|A)$

Example: Find the probabilities of randomly drawing two aces in succession from an ordinary deck of 52 playing cards if we sample

(a) **With** replacement

(b) **Without** replacement

Theorem 2.9 can be generalized.

Theorem 2.10:

If A, B and C are any three events in a sample space S such that $P(A \cap B) \neq 0$

$$P(A \cap B \cap C) = P(A) \cdot P(B|A) \cdot P(C|A \cap B)$$

Proof:

Applications of Conditional Probability

Sensitivity and Specificity

Sensitivity – Ability to detect a true positive

Specificity – Ability to detect true negative

PPV – Positive Predictive Value = Probability that people with a positive screening test result indeed do suffer from the illness.

NPV – Negative Predictive Value = Probability that people with a negative screening test result indeed do not suffer from the illness.

		Truth		
Test Result		Disease	Non Disease	Total
	Positive			
	Negative			
	Total			

Prevalence of
Disease =

Sensitivity =

Specificity =

PPV =

NPV =

2.4 Independent Events

Informally speaking two events A and B are independent if the occurrence or non-occurrence of either one does not affect the probability of the occurrence of the other.

Symbolically, two events A and B are independent if

Each of these equalities implies the other.

It means that if $P(B|A) = P(B)$ and $P(B) \neq 0$ then $P(A|B) = P(A)$

If we substitute $P(B)$ for $P(B/A)$ into the formula, we get

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$P(B) = \frac{P(A \cap B)}{P(A)}$$

Definition: Two events A and B are independent if and only if,

$$P(A \cap B) = P(A) \cdot P(B)$$

Rule of Total Probability and Bayes' Theorem

In many situations the outcome of an experiment depends on what happened in various intermediate stages.

Example: Student may come to university by bus, bike, car, walk.

A student being selected to university under different subject streams.

Theorem:

If the events $B_1, B_2, \dots, B_k$ constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i=1, 2, \dots, k$, then for any event A in S

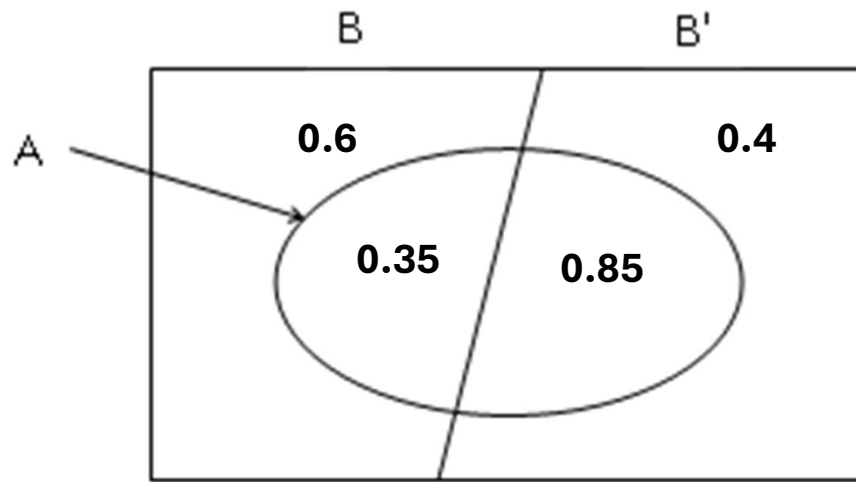
$$P(A) = \sum_{i=1}^k P(B_i) P(A|B_i)$$

This is also known as Rule of Total Probability.

The completion of a construction job may be delayed because of a strike. The probabilities are 0.60 that there will be a strike, 0.85 that construction job will be completed on time if there is no strike, and 0.35 that the construction job will be completed on time if there is a strike. What is the probability that the construction job will be completed on time?

A- Construction job will be completed on time

B- There will be a strike



$$P(A) = (0.6)(0.35) + (0.4)(0.85)$$

Example:

A factory has three production lines, L_1 , L_2 , and L_3 .

40% of the products are manufactured by L_1 ,

35% are manufactured by L_2 , and the remaining are manufactured by L_3 .

The probabilities that a product is defective given each production line are: 0.02 if line 1, 0.05 if line 2 and 0.03 if line 3 was used.

If a product is selected at random from the day's production.

(a) Find the probability that the selected product is defective.

(b) If the selected product is a defect, what is the probability that it came from Line 1?

Bayes' Theorem:

If $B_1, B_2, \dots, B_k$ constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i=1, \dots, k$, then for any event A in S such that $P(A) \neq 0$,

$$P(B_r | A) = \frac{P(B_r) \cdot P(A|B_r)}{\sum_{i=1}^k P(B_i) \cdot P(A|B_i)} \text{ for } r=1, 2, \dots, k$$

Example: A hospital uses two laboratories, L_1 and L_2 , to process blood samples. If Lab 1 was used, the probability that the results will be available on time is 0.75. If Lab 2 was used, the probability that the results will be available on time is 0.85. Suppose that assigning a sample to each lab is equally likely.

If a sample is selected at random from all samples processed that day.

- (a) Find the probability that the selected sample was not available on time.
- (b) Given that the selected sample was not available on time, find the probability that it was processed by laboratory L_2 .
- (c) Are the two event, “Assigning a Lab” and “Getting results on time”, independent? Justify.

Example:

There are four cashiers at a supermarket and a survey was done to measure the satisfaction of customers at check out about the service on these four cashiers. 75%, 80%, 90% and 95% of customers said they are satisfied on the service provided by each cashier A,B,C and D respectively. If selecting a cashier at check out is equally likely for all four cashiers, answer the following questions.

- i) $P(\text{customer is satisfied})$
- ii) $P(\text{customer is not satisfied})$
- iii) $P(A | \text{customer is satisfied})$
- iv) $P(D | \text{customer is not satisfied})$